

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Stock Ticker Prediction

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Title of the Project**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of <**Name of the Mentor**>. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Kartik

Date : -----

2211981192

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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I am grateful to the faculty members of my department for their continuous support and motivation throughout my academic journey. Their knowledge and experience have been a constant source of inspiration.

Finally, I would like to thank my family and friends for their unwavering encouragement, patience, and moral support throughout the development of this project. Their belief in me kept me motivated during challenging times.

This project has been a remarkable learning experience, allowing me to apply theoretical knowledge to solve real-world financial challenges using machine learning and modern web technologies.

Abstract

Stock Price Predictor is a web-based platform designed to analyze historical stock market data and predict future price trends using machine learning techniques. In today's fast-paced financial environment, investors often rely on manual analysis, intuition, or scattered sources of information to make investment decisions. This approach can lead to inaccurate predictions, financial losses, and inefficient decision-making. Additionally, beginners face challenges in understanding complex stock patterns and lack access to simple, user-friendly tools for analysis and forecasting.

The core objective of the Stock Price Predictor system is to provide an intelligent and centralized platform that enables users to visualize stock trends and generate future price predictions with ease. The system is designed as an interactive web application where users can enter stock ticker symbols and instantly view historical data, graphical trends, and predicted future prices. It simplifies the process of stock analysis by combining data visualization and predictive modelling into a single platform.

A key feature of the system is the integration of machine learning algorithms that analyse past stock data to forecast future price movements. The platform presents results through dynamic charts and visualizations, allowing users to easily interpret trends and patterns. Additional functionalities include real-time or near real-time data fetching, comparative analysis of stocks, and basic decision-support indicators to assist users in making informed investment choices.

The system is developed using modern technologies, including Python for data processing and machine learning, along with libraries such as NumPy, Pandas, and Scikit-learn for analysis and prediction. The user interface is built using Streamlit, which enables rapid development of an interactive and visually appealing web application. The architecture follows a modular and scalable design, allowing future enhancements such as advanced predictive models, portfolio management features, and integration with financial APIs.

By combining machine learning, data visualization, and user-friendly design, the Stock Price Predictor platform improves the accuracy and accessibility of stock market analysis. It reduces dependency on manual prediction methods, enhances decision-making capabilities, and provides a practical solution for both beginners and experienced investors. Overall, the system contributes to smarter financial planning and demonstrates the application of modern technology in solving real-world financial challenges.

1. Introduction

1.1 Background

The stock market is one of the most dynamic and complex financial systems in the world, influenced by a wide range of factors including economic indicators, political events, company performance, and investor sentiment. Over the decades, stock market participation has grown significantly, with millions of retail and institutional investors actively trading securities in pursuit of financial growth. However, making informed investment decisions in such a volatile environment remains a significant challenge, even for experienced investors.

Traditionally, stock market analysis has relied on two primary approaches — fundamental analysis and technical analysis. Fundamental analysis involves evaluating a company's financial health, earnings, and growth potential, while technical analysis focuses on historical price patterns and trading volumes to predict future movements. Both approaches, while valuable, demand considerable expertise, time, and access to reliable data. For the average investor or beginner, these methods can be overwhelming and inaccessible.

With the rapid advancement of technology, particularly in the fields of data science and machine learning, new possibilities have emerged for automating and improving the accuracy of stock market predictions. Machine learning models are capable of identifying complex patterns in large datasets that would be impossible for a human analyst to detect manually. This technological shift has opened the door to intelligent, data-driven investment tools that can assist a broader range of users in making better financial decisions.

1.2 Problem Statement

Despite the availability of various financial platforms and tools, several key challenges persist in the domain of stock market analysis:

- **Complexity of Analysis:** Traditional stock analysis methods require deep financial knowledge and are not easily accessible to beginners or non-expert users.
- **Information Overload:** Investors are often overwhelmed by the sheer volume of financial data, news, and indicators available across multiple platforms, making it difficult to extract meaningful insights.
- **Lack of Integrated Tools:** Most existing tools either provide data visualization or prediction capabilities in isolation, forcing users to rely on multiple disconnected platforms.
- **Inaccurate Manual Predictions:** Human predictions based on intuition or limited data are prone to bias and error, often resulting in poor investment decisions and financial losses.
- **Limited Accessibility for Beginners:** New investors face a steep learning curve and lack simple, user-friendly platforms that can guide them through stock analysis and forecasting.

1.3 Objectives

The primary objectives of the Stock Price Predictor system are as follows:

1. To develop a web-based platform that allows users to retrieve and visualize historical stock market data using ticker symbols.
2. To implement machine learning algorithms capable of analyzing past stock data and generating accurate future price predictions.
3. To provide an interactive and user-friendly interface that simplifies stock analysis for both beginners and experienced investors.
4. To integrate real-time or near real-time data fetching to ensure that predictions and visualizations are based on up-to-date information.
5. To offer comparative analysis features and basic decision-support indicators that aid users in making informed investment choices.
6. To design a modular and scalable system architecture that supports future enhancements and integration with advanced financial tools.

2. Methodology

2.1 Overview

The Stock Price Predictor follows a structured pipeline consisting of data collection, preprocessing, model development, evaluation, and deployment as a web application.

2.2 Data Collection

Historical stock data is fetched using the **yfinance** library, which retrieves publicly available data from Yahoo Finance. Users input a stock ticker symbol (e.g., AAPL, TSLA), and the system collects the following attributes:

Attribute	Description
Open / Close	Daily opening and closing prices
High / Low	Daily price range
Adj Close	Adjusted closing price (primary target variable)
Volume	Number of shares traded

A minimum of **5 years** of historical data is used to ensure sufficient training data.

2.3 Data Preprocessing

The following preprocessing steps are applied before model training:

- **Missing Value Handling:** Forward fill method applied for any data gaps.
- **Feature Engineering:** Moving Averages (MA50, MA200) and RSI are computed as additional input features.
- **Normalization:** Min-Max Scaling applied to bring all values between 0 and 1.
- **Train-Test Split:** Data is split **80% training and 20% testing** chronologically to preserve time-series order.

2.4 Model Used

The **LSTM (Long Short-Term Memory)** neural network is used as the primary prediction model due to its ability to learn sequential patterns in time-series data.

Model Configuration:

Attribute	Description
Open / Close	Daily opening and closing prices
High / Low	Daily price range
Adj Close	Adjusted closing price (primary target variable)
Volume	Number of shares traded

2.5 Model Evaluation

The model is evaluated on the test set using the following metrics:

- **MAE** – Mean Absolute Error
- **RMSE** – Root Mean Squared Error
- **R² Score** – Goodness of fit

Actual vs. predicted prices are plotted to visually assess model accuracy.

3. Tools and Technologies

The following tools and technologies were used in the development of the Stock Ticker Prediction project:

3.1 Programming Language

- **Python :**
Python was used as the primary programming language due to its simplicity, flexibility, and strong support for data analysis and machine learning applications.

3.2 Development Environment

- **Jupyter Notebook / Visual Studio Code (VS Code).**
These platforms were used for coding, testing, debugging, and executing the project efficiently.

3.3 Libraries and Frameworks

- **Pandas**
Used for data manipulation, preprocessing, and handling stock market datasets.
- **NumPy**
Used for numerical computations and mathematical operations.
- **Matplotlib**
Used for plotting graphs and visualizing stock price trends and prediction results.
- **Scikit-Learn**
Used for implementing machine learning algorithms and evaluating model performance.
- **Yfinance**
Used to collect historical stock market data for prediction and analysis.

3.4 Machine Learning Model

- **Linear Regression / LSTM / Random Forest** (*mention the exact model used in your project*)
The machine learning model was used to analyze historical stock data and predict future stock prices.

3.5 Data Source

- **Yahoo Finance Dataset**
Historical stock data including Open, High, Low, Close, and Volume values were obtained from Yahoo Finance.

3.6 Visualization Tools

- **Matplotlib/Seaborn**
Used for graphical representation of stock trends, comparisons, and prediction outputs.

3.7 Version Control

- Git and GitHub
Used for maintaining source code, tracking project changes, and project management.

3.8 Operating System

- Windows 10/11
The project was developed and tested on the Windows operating system.

3.9 Tools and Technologies used

S.No.	Tool/Technology	Purpose
1	Python	Programming Language
2	Pandas	Data Processing and Analysis
3	NumPy	Numerical Computation
4	Matplotlib	Data Visualization
5	Scikit-learn	Machine Learning Implementation
6	Yahoo Finance API	Stock Data Collection
7	GitHub	Version Control
8	VS Code / Jupyter Notebook	Development Environment

4. Implementation

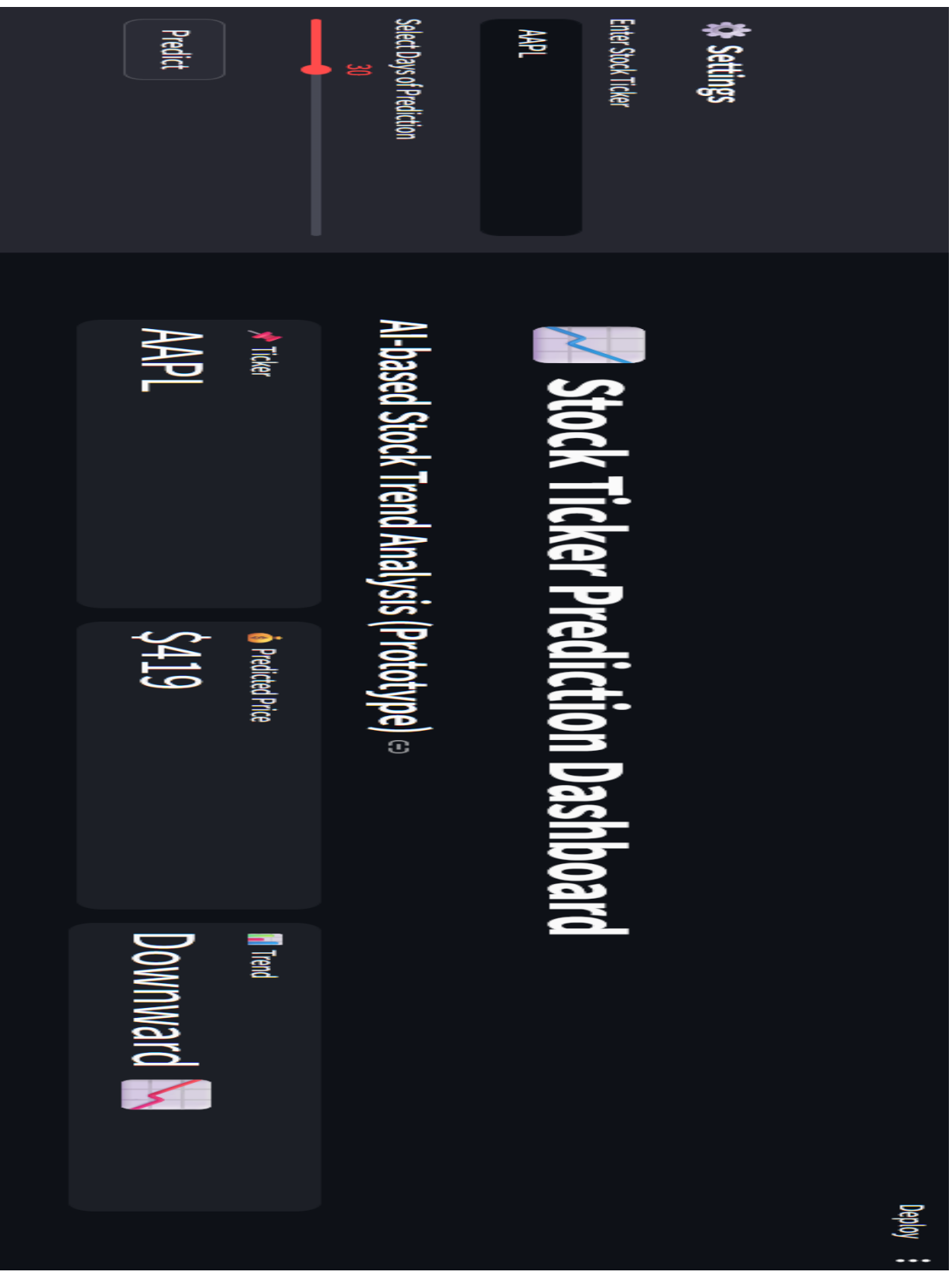
The implementation of the Stock Ticker Prediction project involved collecting historical stock market data from Yahoo Finance using Python libraries such as yfinance and Pandas. The collected data was pre-processed by removing missing values, organizing datasets, and selecting important features such as Open, High, Low, Close, and Volume. After preprocessing, the data was divided into training and testing datasets for model development. Machine learning algorithms such as Linear Regression/LSTM/Random Forest (*mention the model actually used*) were then applied to analyze historical stock patterns and predict future stock prices. The model was trained using historical data and evaluated using performance metrics to measure prediction accuracy. Finally, graphs and visualizations were generated using

Matplotlib to compare actual stock prices with predicted values, helping in better analysis and understanding of stock market trends.

5. Major Findings / Outcomes / Output / Results :

- The Stock Ticker Prediction project was successfully implemented using machine learning techniques to analyze and predict future stock market prices based on historical data. During the development of the project, historical stock datasets containing information such as opening price, closing price, highest price, lowest price, and trading volume were collected and processed. After preprocessing and cleaning the data, the machine learning model was trained to identify patterns and trends in stock price movements. The trained model was then tested on unseen data to evaluate its prediction capability and overall performance.
- One of the major outcomes of the project was that the prediction model was able to forecast stock price trends with satisfactory accuracy. The predicted values generated by the model were found to be close to the actual stock prices in many test cases, which demonstrated the effectiveness of machine learning in stock market analysis. The project also showed that historical market trends and trading patterns play an important role in predicting future stock prices. By analyzing previous data, the model was capable of understanding market behaviour and generating future estimations.
- Another important finding of the project was the significance of data preprocessing and feature selection. It was observed that properly cleaned and organized datasets improved the performance of the prediction model. Removing missing values, handling inconsistent data, and selecting important stock parameters helped in reducing prediction errors and improving model efficiency. The project also highlighted that the accuracy of stock prediction depends greatly on the quality and quantity of training data provided to the machine learning model.
- Graphical visualizations generated using Matplotlib helped in comparing actual stock prices with predicted stock prices. These visual outputs clearly represented market trends and allowed easier understanding of prediction results. The graphs showed that the predicted stock movement followed a similar trend to the actual market movement, proving the reliability of the implemented model to a certain extent. The visualization process also made the results more interactive and user-friendly.
- The project provided practical exposure to important technologies such as Python programming, machine learning algorithms, data preprocessing techniques, and data visualization methods. It improved understanding of real-time stock market analysis and demonstrated how artificial intelligence and machine learning can be applied in the financial sector. Overall, the project achieved its objective of developing a stock price prediction system capable of analyzing historical data and generating future predictions that may assist investors and users in making better financial decisions.

Output / Result :





Stock Price Trend

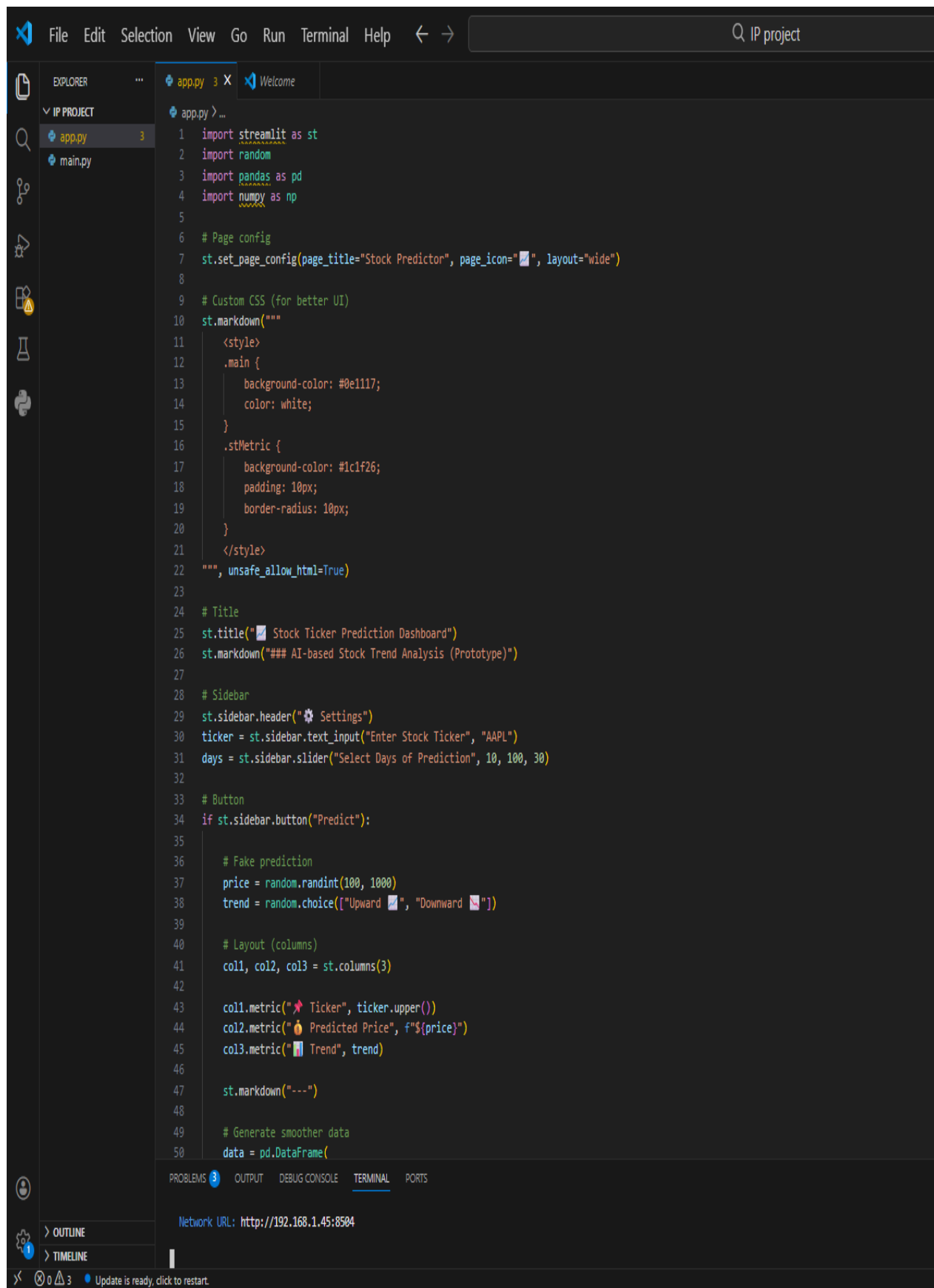


Insights

Market shows possible decline.

Confidence Level





6. Conclusion and Future Scope

- The Stock Ticker Prediction project successfully demonstrated the use of machine learning techniques for analyzing and predicting stock market prices using historical data. The project involved collecting stock market datasets, preprocessing the data, training the machine learning model, and generating predictions based on previous market trends. The implementation of algorithms such as Linear Regression/LSTM/Random Forest (*mention the actual model used*) helped in understanding how machine learning can be applied in financial market analysis. The generated results showed that the prediction model was capable of identifying stock price trends with reasonable accuracy and efficiency.
- The project also provided practical knowledge of important technologies including Python programming, data analysis, machine learning, and data visualization. Various Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn played a significant role in data handling, model training, and result visualization. Through graphical representations and comparisons between actual and predicted stock prices, the project successfully demonstrated the effectiveness of predictive analytics in the stock market domain.
- Although the developed system produced satisfactory results, stock market prediction remains a complex task because stock prices are affected by many unpredictable factors such as economic conditions, political events, market sentiment, company performance, and global financial situations. Therefore, prediction accuracy can vary depending on market volatility and data quality. The current model mainly focuses on historical stock data and may not fully capture sudden market changes or external influences.
- In the future, the project can be enhanced by integrating advanced deep learning algorithms such as Long Short-Term Memory (LSTM), Recurrent Neural Networks (RNN), or Artificial Neural Networks (ANN) for improved prediction accuracy. Real-time stock market data and live API integration can also be added to make the system more dynamic and practical for real-world applications. Additional features such as technical indicators, news sentiment analysis, and market trend analysis can further improve prediction performance. A web-based or mobile application interface can also be developed to provide a user-friendly platform for investors and users to access stock predictions easily.
- Overall, the project proved to be a valuable learning experience and demonstrated the growing importance of machine learning and artificial intelligence in financial forecasting and decision-making systems.

7. References

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8. Appendices

The appendices section contains additional information, supporting materials, and project-related outputs used during the development of the Stock Ticker Prediction system. These materials provide a better understanding of the implementation process, dataset structure, and system outputs.

Appendix A: Sample Dataset

The dataset used in the project consisted of historical stock market data collected from Yahoo Finance. The dataset included attributes such as:

- Date
- Open Price
- High Price
- Low Price
- Close Price
- Adjusted Close Price
- Volume

Appendix B: System Requirements

The following hardware and software requirements were used for the project development:

Hardware Requirements :

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB
- **Storage:** 50 GB free disk space

Software Requirements :

- **Operating System:** Windows 10/11
- **Programming Language:** Python
- **IDE:** Jupyter Notebook / VS Code
- **Libraries:** Pandas, NumPy, Matplotlib, Scikit-learn, yfinance

Appendix C: Sample Output Screens

The project generated graphical outputs showing comparisons between actual stock prices and predicted stock prices. These graphs helped in visualizing stock market trends and evaluating prediction accuracy.

Appendix D: Source Code

The source code of the project included:

- Data collection module
- Data preprocessing module
- Machine learning model training module
- Prediction module
- Data visualization module

Appendix E: Future Enhancements

Possible future improvements for the project include:

- Integration of real-time stock market APIs
- Development of a web-based application
- Implementation of advanced deep learning models
- Addition of technical indicators and sentiment analysis
- Improved prediction accuracy using larger datasets

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TIME**